Core Aptitude Concepts for MCQ Generation

1. Number System

The number system forms the foundation of quantitative aptitude.

Types of numbers:

- **Natural Numbers**: 1, 2, 3, ...
- **Whole Numbers**: 0, 1, 2, 3, ...
- **Integers**: ..., -2, -1, 0, 1, 2, ...
- **Rational Numbers**: Numbers that can be expressed as p/q
- **Irrational Numbers**: Cannot be expressed as p/q (e.g., √2, π)

Important concepts:

- **Prime Numbers**: Numbers divisible only by 1 and itself
- **Composite Numbers**: Numbers with more than two factors
- **LCM (Least Common Multiple)**: Smallest common multiple of numbers
- **HCF/GCD (Highest Common Factor)**: Largest common factor

Divisibility rules examples:

- Divisible by 2 → last digit even
- Divisible by 3 → sum of digits divisible by 3
- Divisible by 5 → last digit 0 or 5
- Divisible by 9 → sum of digits divisible by 9

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2. Percentages

Percentage represents a number as a fraction of 100.

Formula:
Percentage = (Part / Whole) × 100

Key concepts:

- **Increase %** = (Increase / Original value) × 100
- **Decrease %** = (Decrease / Original value) × 100
- **Successive percentage change** uses multiplication factors.

Example:
If a price increases by 20%, the new value becomes 1.2 × original value.

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3. Ratio and Proportion

A ratio compares two quantities.

Example:
a : b

Proportion represents equality of two ratios.

Example:
a / b = c / d

Types:

- **Direct proportion**: Both quantities increase or decrease together.
- **Inverse proportion**: One increases while the other decreases.

Used in problems involving:

- mixture
- partnership
- scaling quantities

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4. Profit and Loss

Important terms:

- **Cost Price (CP)**: price at which an item is bought
- **Selling Price (SP)**: price at which it is sold

Formulas:

Profit = SP − CP
Loss = CP − SP

Profit % = (Profit / CP) × 100
Loss % = (Loss / CP) × 100

If SP > CP → Profit
If SP < CP → Loss

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5. Simple Interest

Interest calculated only on the principal amount.

Formula:

$SI = (P \times R \times T) / 100$

Where:
P = Principal
R = Rate of interest
T = Time

Amount = Principal + Simple Interest

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6. Compound Interest

Interest calculated on principal plus accumulated interest.

Formula:

$A = P(1 + R/100)^T$

Where:
A = Final amount
P = Principal
R = Rate
T = Time

Compound interest = A − P

Used in financial calculations and investment problems.

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7. Time, Speed and Distance

Basic formula:

Speed = Distance / Time
Distance = Speed × Time
Time = Distance / Speed

Key ideas:

- **Relative speed** when two objects move towards or away from each other.
- Used in train, boat, and race problems.

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8. Time and Work

If a person can complete work in N days:

Work per day = 1/N

If two people work together:

Combined work rate = Rate of A + Rate of B

Example:

If A completes work in 10 days
A's rate = 1/10

If B completes work in 15 days
B's rate = 1/15

Combined rate = 1/10 + 1/15

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9. Averages

Average represents the central value of a dataset.

Formula:

Average = Sum of values / Number of values

Example:

Average of 5 numbers = total sum ÷ 5

Types of averages:

- Simple average
- Weighted average

Used in marks, salary, age, and score calculations.

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10. Permutations and Combinations

Permutation

Arrangement where **order matters**.

Formula:

nPr = n! / (n − r)!

Example:
Arranging students in positions.

Combination

Selection where **order does not matter**.

Formula:

$nCr = n! / (r! (n - r)!)$

Example:
Selecting students for a team.

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11. Probability

Probability measures the chance of an event occurring.

Formula:

$P(E) = \text{Favorable outcomes} / \text{Total outcomes}$

Range:

$0 \leq \text{Probability} \leq 1$

Concepts:

- **Certain event** → probability = 1
- **Impossible event** → probability = 0
- **Mutually exclusive events** cannot occur together.

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12. Logical Reasoning Basics

Common reasoning question types:

- **Number Series**
  Find the pattern in a sequence.
- **Analogy**
  Relationship between two pairs.
- **Odd One Out**
  Identify the item that does not belong.
- **Coding–Decoding**
  Letters or numbers are coded based on patterns.

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13. Data Interpretation

Questions based on data representation.

Common formats:

- Tables
- Bar graphs
- Pie charts
- Line graphs

Skills required:

- quick percentage calculations
- ratio comparisons
- identifying trends

Data interpretation tests analytical and numerical reasoning skills.